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How AI and IoT Is Transforming Digital Payment Systems: A Modernization Blueprint for Banks

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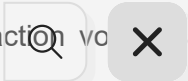
Neeraj Aggarwal Senior Project Manager Infosys Limited

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By Neeraj Aggarwal

Digital payments have quietly become one of the most complex, high-stakes systems in modern banking. Every tap, swipe, or API call triggers a chain of events involving fraud checks, compliance rules, routing decisions, settlement processes, and real-time risk scoring. As transaction vo



surge and customer expectations rise, legacy payment infrastructures—many built decades ago—are struggling to keep pace.

Artificial intelligence is emerging as the most powerful modernization lever available to banks today. It is not just improving fraud detection or automating manual tasks; it is reshaping the architecture, resilience, and operating model of digital payment systems. This article presents a practical blueprint for how banks can use AI to modernize their payment infrastructure, drawing on real-world modernization patterns and proven frameworks.

1. The Hidden Complexity Behind Every Digital Payment

To most customers, a payment is a simple action: tap a card, scan a QR code, or click “Pay Now.” But behind the scenes, a single transaction may involve:

- Identity verification
- Device fingerprinting
- Fraud scoring
- Routing across multiple networks
- Authorization and clearing
- Settlement and reconciliation
- Logging, monitoring, and compliance checks

Legacy systems were never designed for this level of real-time intelligence. They were built for batch processing, predictable volumes, and static rules. Today’s environment—instant payments, cross-border flows, 24/7 uptime, and sophisticated fraud—demands a fundamentally different approach.

AI provides that shift.

2. Where AI Creates the Most Impact in Digital Payments

AI is transforming digital payments across four major dimensions: **fraud prevention**, **operational efficiency**, **modernization**, and **customer experience**.

a. AI for Fraud Detection and Risk Scoring

Traditional fraud systems rely on static rules that attackers quickly learn to bypass. AI changes the game by:

- Detecting anomalies in real time



- Learning behavioral patterns
- Adapting to new fraud vectors
- Scoring risk dynamically based on context

Banks using AI-based fraud models report significant reductions in false positives and faster detection of emerging threats.

b. AI for Operational Efficiency

Payments generate enormous operational overhead—exceptions, disputes, reconciliation mismatches, and routing failures. AI can:

- Predict the best routing path
- Automate exception handling
- Identify root causes of failures
- Optimize settlement cycles
- Reduce manual interventions

This is especially valuable in high-volume systems where even a 1% efficiency gain translates into millions saved.

c. AI for Modernization and Legacy Transformation

This is where your modernization frameworks come into play.

ERMI (Efficiency, Resilience, Modernization, Integration)

AI enhances each dimension:

- **Efficiency:** Automated testing, synthetic data, intelligent routing
- **Resilience:** Predictive failure detection, anomaly monitoring
- **Modernization:** AI-driven code analysis, dependency mapping
- **Integration:** Smart API orchestration, event-driven workflows

LNBA (Low-Noise Banking Architecture)

AI helps banks modernize without disrupting mission-critical systems by:

- Identifying low-risk modernization pathways
- Automating regression validation
- Supporting coexistence between legacy and modern components

AI-Driven Validation Framework



AI can generate test cases, simulate transaction flows, and validate system behavior at scale—critical for regulated payment systems.

d. AI for Customer Experience

AI enhances the customer journey through:

- Personalized payment flows
- Intelligent dispute resolution
- Conversational interfaces for support
- Real-time insights into spending patterns

Payments become not just faster, but smarter and more intuitive.

3. The Role of IoT and Edge Intelligence in Payments

As IoT devices proliferate—wearables, connected cars, smart POS terminals—payments are moving closer to the edge. AI at the edge enables:

- Low-latency fraud detection
- Device-level risk scoring
- Local anomaly detection
- Offline-capable transaction validation

This aligns with the White House CET categories such as **Integrated Communication and Networking Technologies**, **AI**, and **Digital Payment Technologies**.

IoT-driven telemetry also enhances observability, giving banks real-time visibility into device health, transaction flows, and network performance.

4. A Modernization Blueprint for Banks

Here is a practical, step-by-step blueprint for banks looking to modernize their payment systems using AI.

Step 1: Assess Legacy Constraints Using ERMI

- Identify resilience gaps
- Map failure modes
- Evaluate modernization readiness
- Prioritize high-impact, low-risk improvements



ERMI provides a structured way to understand where AI can deliver the most value.

Step 2: Apply LNBA to Sequence Modernization Work

LNBA helps banks:

- Reduce modernization noise
- Avoid destabilizing core systems
- Introduce AI components incrementally
- Maintain regulatory compliance

This is essential for mission-critical payment systems.

Step 3: Introduce AI-Driven Validation

AI-based validation accelerates modernization by:

- Generating synthetic data
- Automating regression testing
- Simulating high-volume transaction flows
- Detecting edge-case failures

This reduces risk and speeds up deployment cycles.

Step 4: Build an AI-Ready Architecture

Modern payment systems require:

- Event-driven architectures
- API-first design
- Real-time data pipelines
- Cloud-native components
- Observability and telemetry

AI thrives in environments where data flows freely and systems are modular.

Step 5: Operationalize AI Safely

Responsible AI is critical in regulated environments. Banks must implement:

- Model governance
- Bias detection
- Explainability



- Continuous monitoring
- Human-in-the-loop controls

This ensures AI enhances trust rather than undermining it.

5. What a Modernized Payment System Looks Like

A fully modernized, AI-enabled payment system includes:

- **Real-time fraud scoring** at every step
- **Predictive routing** for optimal performance
- **IoT-driven telemetry** for device and network health
- **Automated exception handling**
- **Sub-second transaction latency**
- **Self-healing capabilities** powered by AI

This is not a futuristic vision—it is achievable today with the right architecture and frameworks.

6. Why AI-Driven Modernization Is Now a Strategic Imperative

Banks that modernize their payment systems with AI will:

- Reduce fraud losses
- Improve customer trust
- Lower operational costs
- Increase resilience
- Accelerate innovation
- Compete effectively with fintechs

The White House's Critical and Emerging Technologies (CET) list reinforces the strategic importance of AI, digital payment technologies, and integrated communication systems. Payment modernization is no longer just a technology initiative—it is a national-priority capability.

Conclusion: The Future of Intelligent Payments

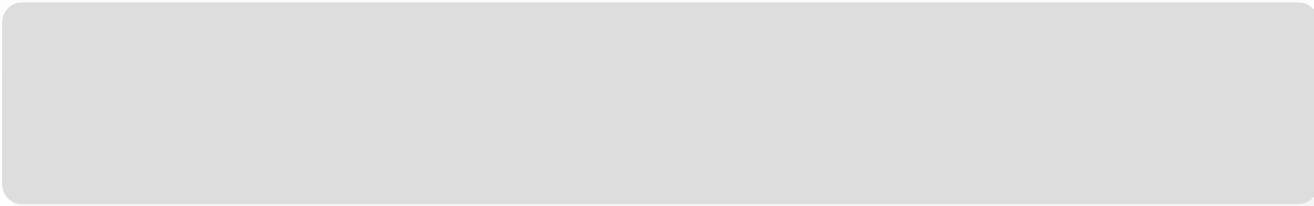
AI is transforming digital payments from a static, rules-based process into an intelligent, adaptive, and resilient ecosystem. Banks that embrace AI-driven modernization will lead the next decade of financial innovation. Those that delay risk falling behind in a world where speed, security, and intelligence define competitive advantage.



The blueprint is clear. The technology is ready. The opportunity is now.

About Author

Neeraj Aggarwal is a recognized modernization leader with deep expertise in AI, core banking, and legacy transformation. He advises financial institutions on building resilient, data-driven architectures and contributes thought leadership across industry forums and publications.



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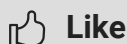
This is one of the few pieces on AI- and IoT-driven payment modernization that actually treats the subject with architectural depth rather than surface-level



futurism. What stands out here is the clear articulation of *how* modernization should be sequenced in regulated banking environments—especially the emphasis on resilience modeling, low-noise modernization, and AI-driven validation. These are the kinds of frameworks banks desperately need but rarely see documented publicly.

The integration of IoT telemetry with real-time fraud scoring and intelligent routing is particularly timely. Most institutions are still treating IoT endpoints as peripheral devices rather than high-value data sources that can materially improve risk posture and customer experience. The blueprint outlined here bridges that gap by showing how event-driven architectures, edge intelligence, and AI-based decisioning can coexist within compliance-bound payment ecosystems.

What's most compelling is that the article doesn't just describe emerging capabilities—it provides a modernization *methodology*. The industry has plenty of technology, but very few structured approaches that help banks move from legacy constraints to AI-ready operating models without destabilizing mission-critical systems. This contribution fills that gap and deserves wider discussion among modernization leaders, regulators, and payment architects.



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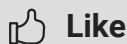


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**A Finextra member**

24 February 2026

The real bottleneck isn't AI or IoT adoption—it's the lack of architectural transparency in legacy payment cores. Until banks modernize lineage, decision-logic traceability, and real-time observability, AI-driven payment intelligence will remain fragile and regulator-sensitive. Modernization isn't a technology upgrade; it's a governance upgrade.



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Report

**Neeraj Aggarwal** Senior Project Manager at Infosys Limited**Author**

25 February 2026

Thank you for the engagement. One aspect I'd like to explore further with practitioners is how banks are approaching the convergence of IoT-scale data, AI-driven decisioning, and real-time settlement. Most organizations are still optimizing individual components, but the real breakthrough comes when payments, risk, and telemetry operate as a unified, explainable system. I'd be very interested in hearing how others are tackling this integration challenge in their modernization programs.



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